

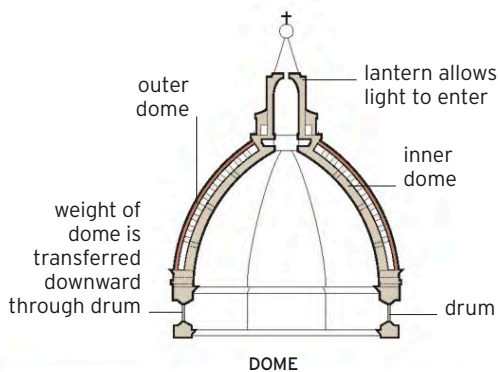
▷ TOWERING CATHEDRAL

Built between 1220 and 1270, Amiens is the tallest complete cathedral in France and has the greatest interior volume. The building is famous for its highly ornate exterior, built using locally quarried stone.

these forms, but they developed them to create a structural system that was not substantially improved upon until the advent of construction in iron and steel. It was the Romans who truly developed the arch, not only in monumental fashion, but also in revolutionary public works. The Romans were the first major bridge and aqueduct builders, employing multiple arches to span rivers and valleys. They were also early proponents of concrete, utilizing it most famously in the glorious dome of the Pantheon (see pp.104–105), which, with a diameter of 142 ft (43.2 m), remained unsurpassed until the 19th century.

Attaining height

Perhaps the most spectacular building achievement of the Classical age—if the descriptions are to be believed—was the Pharos of Alexandria, a great lighthouse built by the Greco-Romans in Egypt in the 3rd century BCE. Perhaps as tall as 330 ft (100 m), it was the first high-rise building. It was not until the 14th century that buildings of a comparable height were rising in Europe again. This came as a result of a boom in cathedral building—a “cathedral crusade.” Sponsored by church and state, medieval masons pushed stone building techniques to even greater heights. The use of trunklike piers, buttresses, and Gothic arches and vaults raised the roofs to the heavens, creating soaring internal spaces, echoed on the outside by towers, steeples, and spires that strained skyward. In what is now Central and South America, pre-Columbian civilizations also employed stone to create sophisticated architectural wonders used for a variety of purposes, including their own version of the stepped pyramid.



△ DOMES

The arch was swiveled through 360 degrees to give us the dome. Developed by the Romans, it reached its apex with Filippo Brunelleschi's dome for Florence Cathedral (see pp.140–141), completed in 1461 and the largest brick dome ever built.

